

Annexure E – Details of the two use cases to be refined with GPSSA during the implementation stage

Use-case#1

Use Case: Analyzing Customer Complaints and Suggestions Using Gen AI

Overview:

This use case involves an organization implementing a system that leverages Generative AI (Gen AI) to automatically analyze customer complaints and suggestions submitted through various channels (e.g., email, social media, website forms). The goal is to efficiently categorize, prioritize, and assign actions to the appropriate teams, ensuring faster and more effective responses.

Key Process Steps:

1. Complaint and Suggestion Collection:

- Customers submit their complaints and suggestions via different platforms (email, web forms, phone calls, social media).
- The system aggregates all incoming customer feedback into a central "queue."

2. AI-Driven Content Understanding:

- **Gen AI** processes the incoming feedback and understands the sentiment, intent, and subject matter of each message. It categorizes them into:
 - **Complaints** (e.g., service issues, product defects)
 - **Suggestions** (e.g., improvements, feature requests)
 - **Neutral Feedback** (e.g., general inquiries, praise)
- The AI uses natural language processing (NLP) to identify keywords, phrases, and patterns, assessing urgency, severity, and specific topics (e.g., "late delivery," "refund request," "more payment options").

3. Queue Management:

- The AI system organizes complaints and suggestions into a dynamic queue.
- Each item in the queue is tagged with:
 - **Priority Level** (e.g., High, Medium, Low based on urgency)

- **Category** (Complaint, Suggestion, Inquiry)
- **Assigned Team** (e.g., pension sector, IT, etc)

4. Response Generation:

- **Gen AI** generates personalized responses for each complaint or suggestion, based on pre-defined templates or custom responses:
 - **Complaint Responses:** Automated responses that acknowledge the issue, offer troubleshooting steps, or escalate the case to an appropriate agent.
 - **Suggestion Responses:** Responses that thank the customer for their input and outline any potential actions or timelines for incorporating their suggestions.
 - **Neutral Feedback Responses:** Acknowledgement and appreciation for feedback.

5. Action Assignment:

- Based on the AI's understanding of the content, the system assigns the appropriate action to the relevant team.
 - **Complaint Escalation:** If a complaint is categorized as high priority or urgent (e.g., a payment issue), the system assigns it to a customer support agent immediately.
 - **Product/Feature Suggestions:** Suggestions related to product development are forwarded to the product management or engineering team.
 - **Routine Inquiries:** General inquiries are sent to the support team or a self-service knowledge base.
- The queue is prioritized according to the urgency and severity of the issues.

6. Work Division and Team Collaboration:

- **Support Team:** Handles urgent and critical complaints, ensuring fast resolutions for time-sensitive issues.
- **Product/Engineering Teams:** Focus on complaints related to product functionality or design, while considering customer suggestions for future improvements or feature requests.
- **Marketing/Communication Teams:** Address inquiries or provide updates on new features or company policies based on customer feedback.

The AI system also integrates with existing tools (CRM, helpdesk, project management software) to assign tasks automatically to the right team members and track progress in real-time.

Response and Actions:

1. Immediate Automated Response:

- Upon receiving customer feedback, Gen AI can send an instant acknowledgment. For complaints, it might say:
 - *“Thank you for contacting us. We’re sorry to hear about your issue. Our team is reviewing your concern and will get back to you shortly.”*
- For suggestions:
 - *“We appreciate your suggestion! Our team will review it for possible future improvements. Thank you for helping us get better.”*

2. Complaint Resolution Action:

- **High-Priority Complaints** (e.g., payment issues, major bugs) will trigger immediate escalation to a live agent.
- **Medium-Priority Complaints** (e.g., late delivery, minor product defects) may be handled through automated troubleshooting steps or transferred to a support team for further action.
- **Low-Priority Complaints** (e.g., simple queries, informational requests) will be addressed with a combination of automated responses and knowledge base links.

3. Action Plan for Suggestions:

- The system records customer suggestions and either triggers a review by the product team or provides an automated update if the suggestion is being considered for future releases.
- For example:
 - *“Thank you for your suggestion about adding a new feature. Our product team reviews all feature requests during each development cycle, and we’ll notify you if it gets implemented!”*

4. Regular Updates:

- Automated follow-ups can be sent to customers, keeping them informed about the status of their issue, whether it’s being resolved or if their suggestion is under review.

- For ongoing or unresolved complaints, the system schedules periodic reminders or escalations until resolution.

5. **Workload Management:**

- The AI helps ensure that workload distribution is balanced across teams. For instance, if there are too many high-priority tickets, AI can help identify underutilized resources and reassign tasks to optimize throughput.
- It ensures no tickets are left unaddressed by alerting team members about pending cases.

Benefits of Using Gen AI:

1. **Faster Response Times:** Automated categorization and response generation reduce the time to acknowledge and resolve customer issues.
2. **Personalized Interactions:** AI-generated responses maintain a human-like tone, improving customer satisfaction.
3. **Efficient Resource Allocation:** AI streamlines task assignment based on issue severity and team expertise, optimizing resource utilization.
4. **Scalable Support:** The system can scale with growing customer feedback volumes without needing a proportional increase in staff.
5. **Continuous Learning:** AI models learn from customer interactions, allowing them to improve over time and generate better responses and insights.

This approach creates an efficient system that handles customer complaints and suggestions in a more organized and effective way, while also enhancing the customer experience through timely and thoughtful responses.

Use-case#2

Use Case: AI-Driven Data Scanning and Insight Generation in a Data Lake

Overview:

The use case focuses on using Artificial Intelligence (AI) to scan and analyze large volumes of data in a Data Lake, building actionable insights and linking value based on the actual data. The AI system identifies data quality issues, inconsistencies, and anomalies, and then suggests potential corrections along with an assessment of the impact on business outcomes.

Business Challenge:

Organizations often store large, disparate datasets in Data Lakes, but these datasets can be messy, inconsistent, and unstructured. Without effective data governance and quality control mechanisms, it becomes challenging to extract actionable insights, leading to poor decision-making, missed opportunities, and business inefficiencies. The challenge is to leverage AI to automate the process of identifying these data issues and provide real-time insights that link directly to business value.

Key Process Steps:

1. **Data Ingestion:**
 - Raw data is ingested into the Data Lake from various sources
2. **AI-Powered Data Scanning:**
 - **AI models** scan the Data Lake and process the ingested data, identifying patterns, correlations, and anomalies across various datasets. The AI identifies:
 - **Data Quality Issues:** Missing values, outliers, duplicates, and errors in the data.
 - **Inconsistencies:** Conflicting data across different data sources (e.g., mismatched between EBS data and PAP).
 - **Anomalies:** Unexpected data spikes or drops that might indicate fraud, system errors, or operational inefficiencies.
3. **Data Value Linkage and Insight Building:**

- AI correlates the data with business outcomes, such as contributions, customer satisfaction, or payroll. By leveraging **predictive analytics** and **machine learning models**, AI identifies key data points that directly impact the business, such as:
 - Predicting contribu.
 - Identifying bottlenecks in the supply chain from inventory and logistics data.
 - Predicting demand forecasting trends based on sales history and external factors like weather or economic conditions.
- AI establishes **value linkage** by connecting data insights to specific business KPIs (e.g., "This inconsistency in customer data could impact our marketing campaign effectiveness by 20%").
- 4. **Data Quality Issue Detection and Impact Assessment:**
 - The AI performs continuous scans for data quality issues and assigns a **severity score** to each issue based on its potential impact on business decisions:
 - **Low Severity:** Minor inconsistencies (e.g., missing descriptions) with little impact on business insights.
 - **Medium Severity:** Missing values or duplicates that can cause minor misinterpretations in reports or dashboards.
 - **High Severity:** Major data errors that could significantly mislead business decisions (e.g., incorrect financial data or inconsistent customer information).
- 5. **Suggested Corrections:**
 - **AI-driven recommendations** are provided to data engineers or analysts for each identified data issue. These might include:
 - **Missing Data:** AI suggests methods for imputation (e.g., using the mean/median, or AI-driven predictions to fill in gaps).
 - **Duplicate Data:** AI highlights rows or records that are duplicates and proposes automatic removal or merging.
 - **Inconsistencies:** AI identifies mismatched data points (e.g., inconsistent customer IDs across sales and support data) and proposes linking mechanisms or automatic standardization rules.
 - **Anomalies:** For detected anomalies, AI proposes potential causes, such as technical glitches or external factors (e.g., holiday spikes in e-commerce data) and possible corrective actions.
- 6. **Impact Analysis:**
 - **Impact Simulation:** Once corrections are suggested, AI models simulate the potential impact of those changes on business operations or financial outcomes. For example:
 - Correcting erroneous sales data might improve revenue forecasting accuracy by 10%.
 - Standardizing customer data across multiple systems could lead to a 15% improvement in marketing campaign targeting.
 - The AI provides a **prioritization** of the suggested corrections based on the potential business impact, helping teams decide where to focus efforts.

7. Automation and Continuous Monitoring:

- The AI system continues to scan incoming data, ensuring that issues are flagged in real-time as new data is ingested into the Data Lake.
- It also continuously learns from the corrections and insights, improving the accuracy of its predictions and recommendations over time.

Tools and Technologies Involved:

- **Data Lake Platforms:** AWS S3, Azure Data Lake, or Google Cloud Storage for storing large volumes of structured and unstructured data.
- **AI and Machine Learning Frameworks:** TensorFlow, PyTorch, or custom-built models for anomaly detection, predictive analytics, and natural language processing.
- **Data Quality Tools:** Talend, Informatica, or customized scripts to automate data cleansing tasks.
- **Data Governance:** Tools like Alation or Collibra to ensure that data quality is maintained across the data lifecycle.

Response and Action Plan:

1. Immediate Data Quality Alerts:

- When an issue is detected (e.g., missing data in a critical financial dataset), AI sends **automated alerts** to the data quality team, along with the suggested corrections and impact assessment.

2. Continuous Monitoring:

- Data engineers receive a **real-time dashboard** showing the status of the data quality, along with actionable insights and the impact of any new changes or anomalies.

3. Proactive Issue Resolution:

- As data corrections are made, AI constantly updates the data quality scorecard, tracking whether the issue was resolved effectively or if further adjustments are needed.

4. Impact Reporting:

- Regular reports summarize the **business value** gained by addressing these issues (e.g., improved decision-making, reduced operational inefficiencies).
- For instance, after correcting the data inconsistency between sales and inventory, the AI reports an improved inventory management system that leads to a 5% reduction in stockouts.

Benefits of Using AI in This Use Case:

- **Improved Data Quality:** AI detects and corrects data issues at scale, ensuring that data used for business decision-making is clean and reliable.
- **Actionable Insights:** AI translates raw data into meaningful business insights, showing exactly how data quality affects KPIs and providing direction for corrective actions.

- **Increased Efficiency:** By automating the identification of data issues and suggesting corrections, organizations can save time and resources compared to manual data management efforts.
- **Data-Driven Decisions:** With improved data quality and accurate insights, business leaders can make more informed, data-driven decisions, leading to better operational outcomes.
- **Scalability:** AI enables organizations to scale their data operations and handle increasing volumes of data without overwhelming human resources.

This approach provides an intelligent, automated system for managing data quality issues in a Data Lake while linking data improvements directly to business value, making it easier to make informed decisions and drive business success.